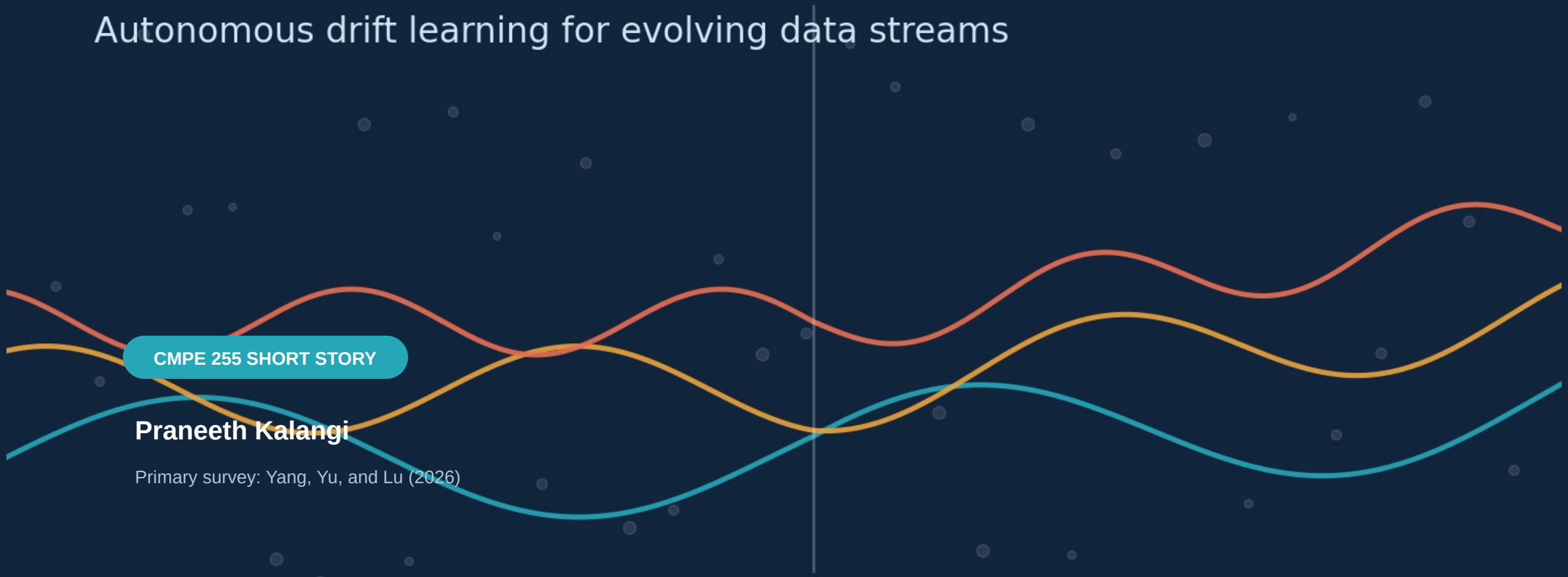
WHEN DATA CHANGES

DRIFT

Autonomous drift learning for evolving data streams



CMPE 255 SHORT STORY

Praneeth Kalangi

Primary survey: Yang, Yu, and Lu (2026)

Time • Data • Model

A production model is a snapshot of a moving process

JANUARY

A fraud model learns yesterday's behavior.

New payment tools, merchant behavior, and attack strategies arrive after deployment.

92%

offline accuracy

does not guarantee

future reliability

$P_t(X,Y)$ changes while the code stays the same

The research question has expanded

From detecting external change to managing an evolving learning system

DETECT

Is the stream statistically different from the recent past?

+

UNDERSTAND

What changed: timing, representation, meaning, or model state?

+

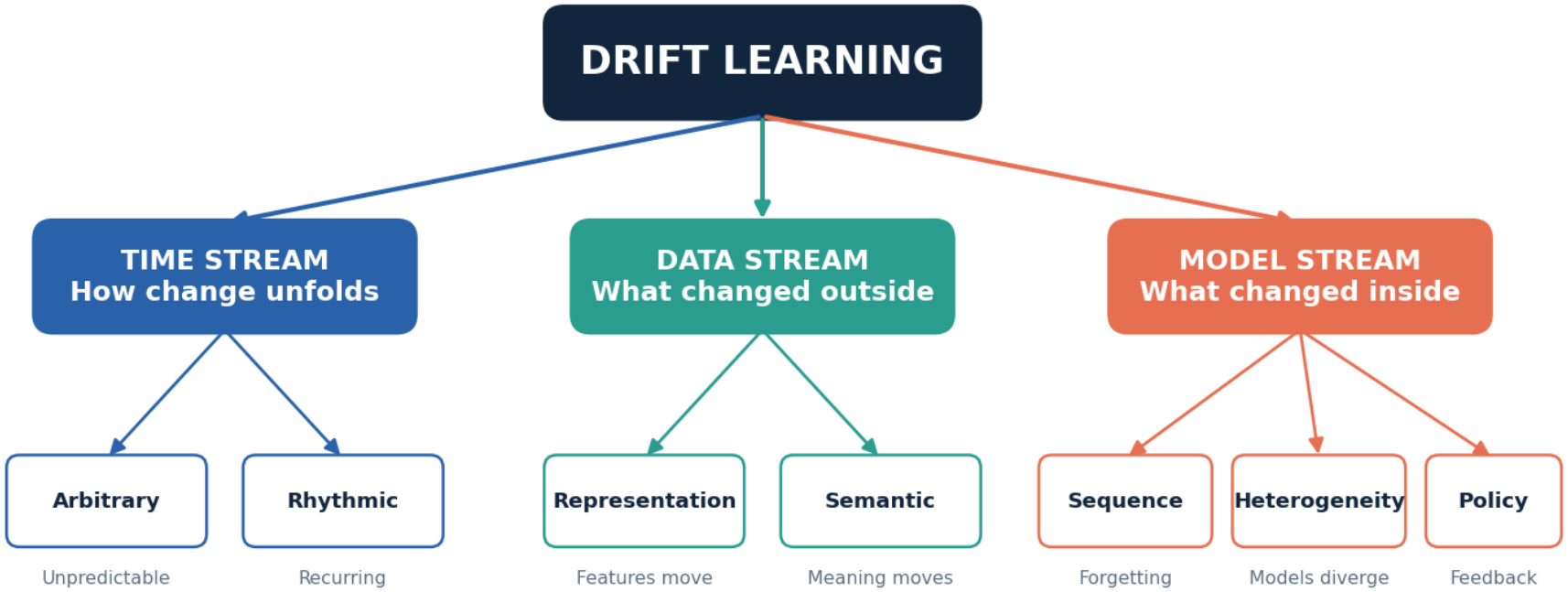
ADAPT

Which response restores performance without causing new damage?

193 studies synthesized into a three-dimensional drift-learning framework

Three dimensions explain where drift originates

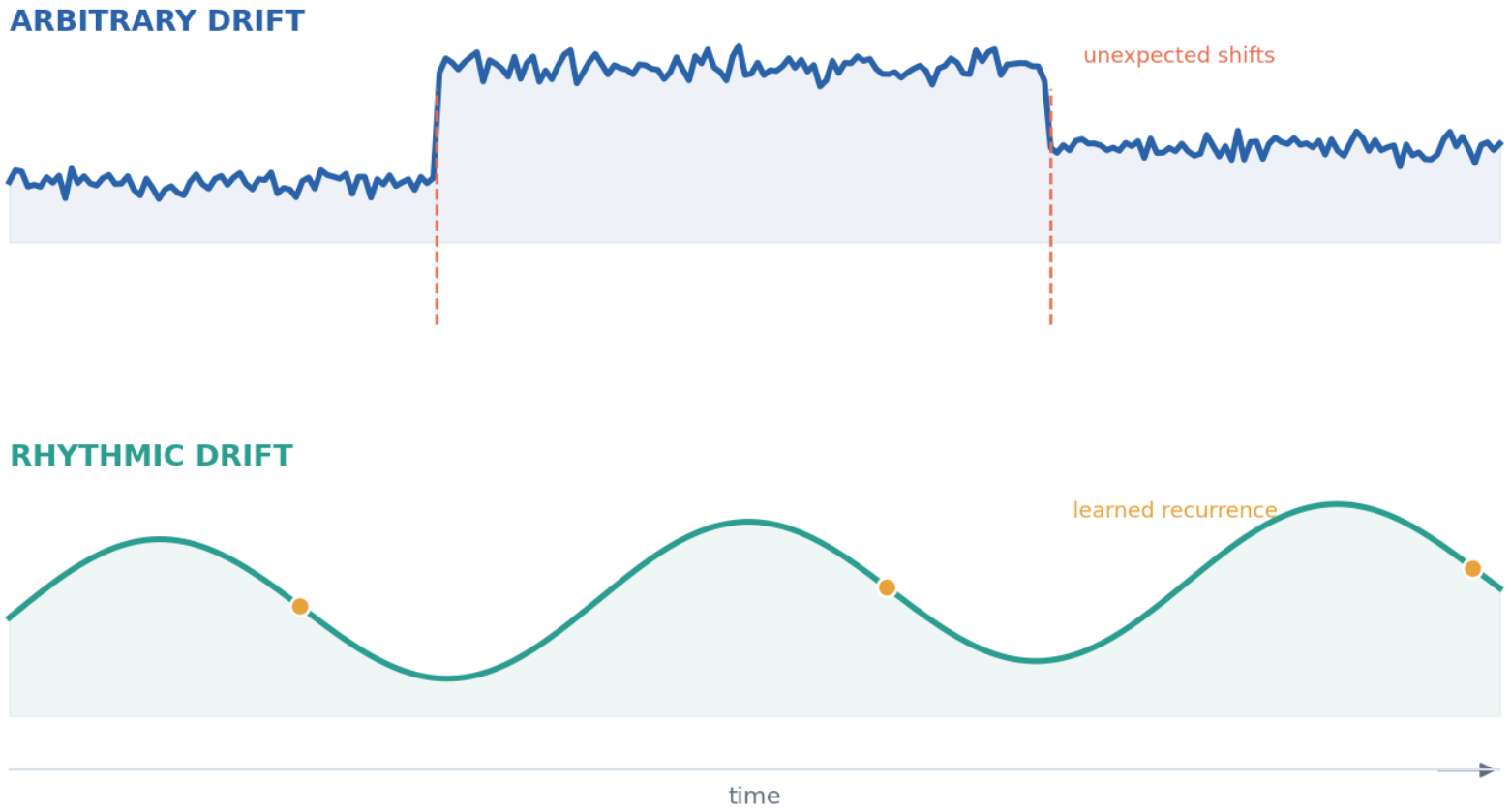
A three-dimensional view of drift learning



The dimensions interact: external change can trigger internal model change, and model actions can reshape future data.

Temporal structure changes the response

Arbitrary drift reacts; rhythmic drift can anticipate



Arbitrary drift needs fast reaction. Rhythmic drift creates a chance to forecast the next regime.

Representation and meaning can move independently

REPRESENTATION DRIFT

The feature space changes

Examples

New camera
Sensor recalibration
Network protocol update

SEMANTIC DRIFT

The input-label relationship changes

Examples

Evolving fraud behavior
New label policy
Changing customer intent

Feature-only monitoring can miss a label-only change

The learner can drift even when incoming data looks stable

SEQUENCE

Continual updates erase earlier knowledge

catastrophic forgetting

HETEROGENEITY

Distributed models gradually diverge

federated clients

POLICY

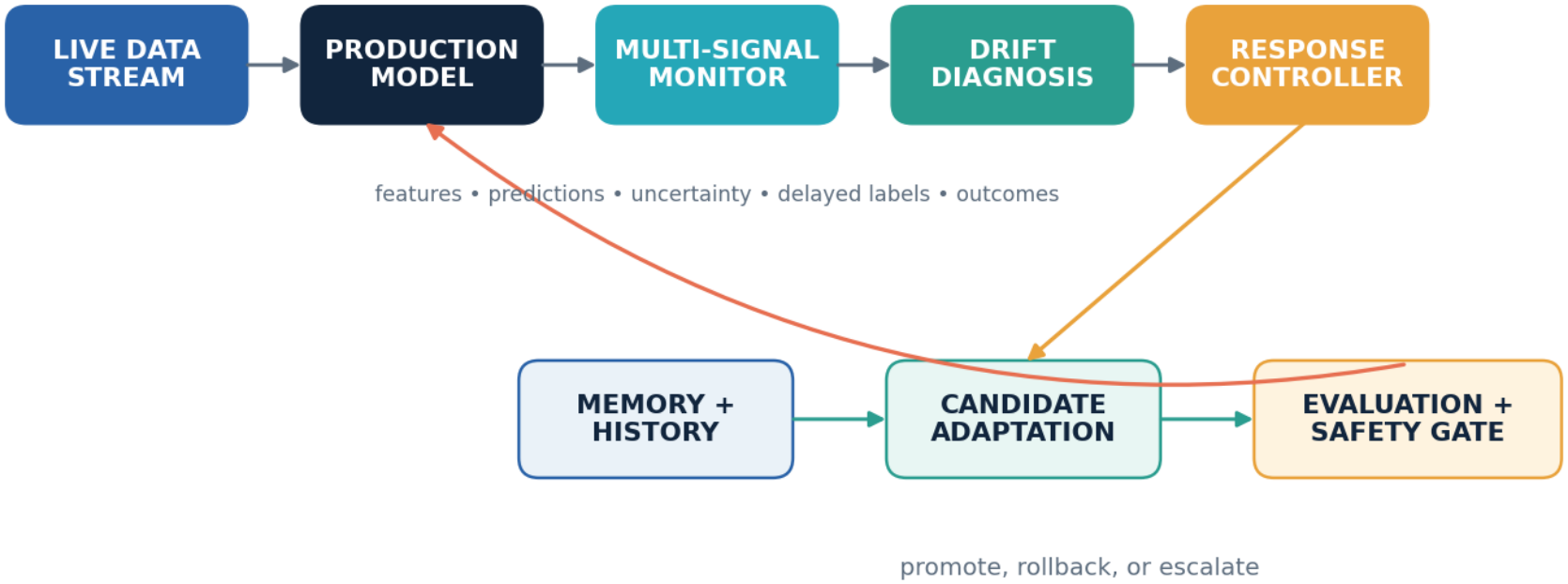
Model actions reshape future observations

feedback loops

Monitor model state, disagreement, and update behavior, not only input features.

Autonomous adaptation needs diagnosis and a safety gate

Reference architecture for controlled drift adaptation



Known drift points make detector evaluation reproducible

A reproducible evaluation pipeline for drift detectors



Four drift mechanisms × abrupt and gradual transitions

class prior | label swap | feature permutation | feature filtering

Precision • Episode recall • F1 detection • Normalized detection time • False-alarm rate

A detector must be accurate, timely, and operationally useful

PRECISION

How many alarms were correct?

EPISODE RECALL

How many drift episodes were found?

F1 DETECTION

How well are precision and recall balanced?

NORMALIZED DELAY

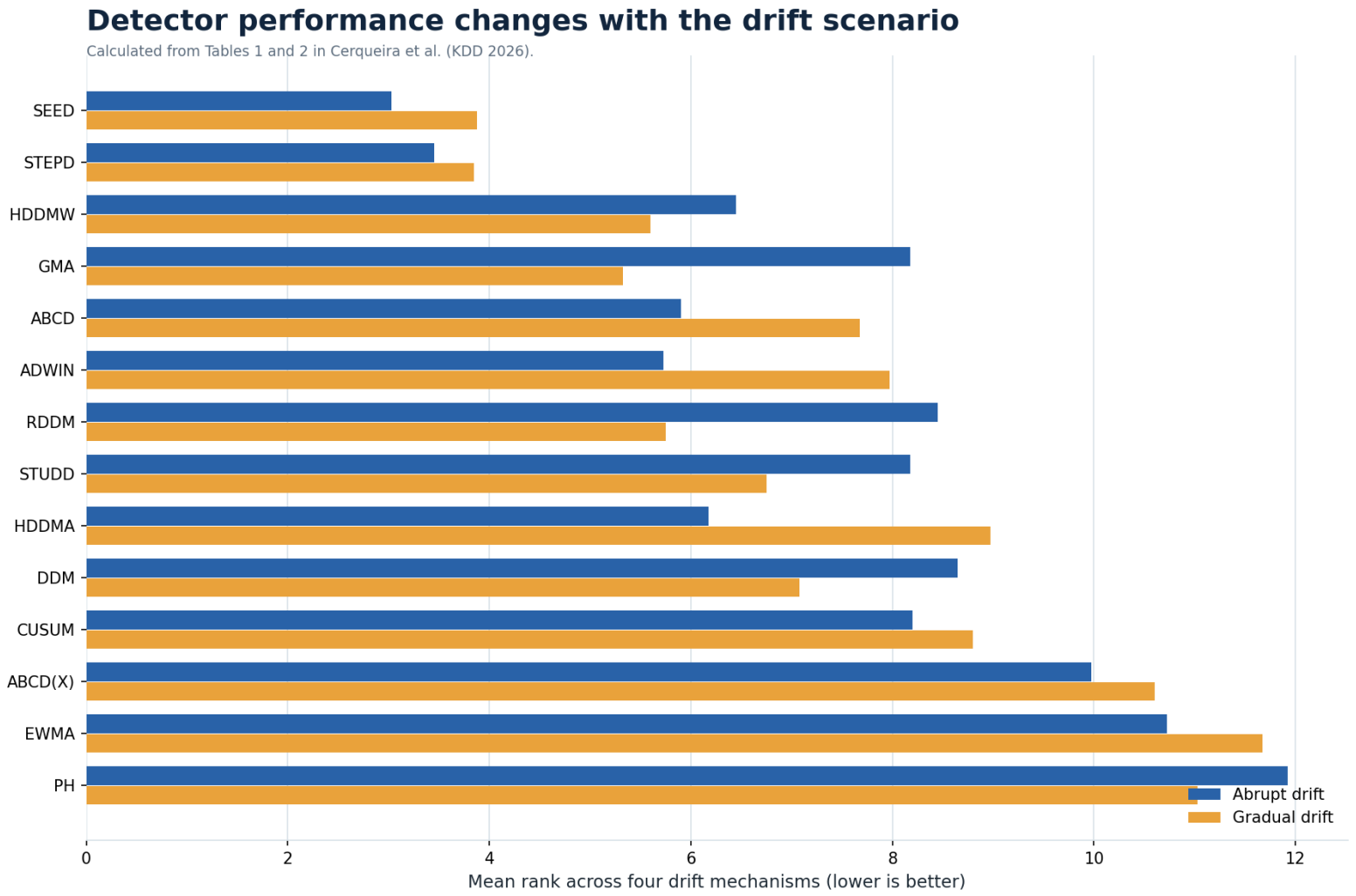
How quickly did the correct alarm arrive?

FALSE-ALARM RATE

How much unnecessary work did alarms create?

No single number captures detection quality and business cost.

Detector performance depends on the drift mechanism



READING THE RESULT

SEED and STEPD stay competitive across many scenarios.

Abrupt drift is usually easier than gradual drift.

Feature-only detectors struggle with label-only change.

No universal winner

Cross-dataset tuning changes the benchmark conclusion

Ablation: tuned configurations versus defaults

DEFAULT CONFIGURATION

Parameters may reflect the datasets used when the detector was originally developed.

Risk: an optimistic or unfair comparison

VS

LEAVE-ONE-DATASET-OUT

Tune on six datasets, then evaluate once on the held-out seventh dataset.

Result: stronger and more transferable settings

The benchmark reports meaningful gains from optimization over defaults.

The same drift problem creates different risks by domain

FRAUD

Attackers adapt to the detector

HEALTHCARE

Population and clinical practice change

RECOMMENDERS

Recommendations reshape user behavior

ROBOTICS

Weather, surfaces, and people change

FINAL TAKEAWAY

The goal is not a model that changes whenever the data moves.

The goal is a system that identifies the type of change, chooses a proportional response, and validates that response before deployment.

TIME

How change unfolds

DATA

What changed outside

MODEL

What changed inside